BSIT 375: Knowledge Discovery and Data Science

Week 4 Lecture Notes: Association Rules

Based on Larose & Larose: *Discovering Knowledge in Data (2nd Ed)*

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1 Affinity Analysis and Market Basket Analysis

Affinity analysis is a data mining technique that discovers co-occurrence relationships among items or events. Its most famous application is **Market Basket Analysis** (MBA), where retailers analyze transaction data to find which products are frequently purchased together. The discovered patterns can be used for:

- **Product placement**: placing complementary items close to each other.
- **Cross-selling**: suggesting additional items to customers.
- **Promotional design**: creating bundles or discounts.
- **Inventory management**: understanding item dependencies.

Example: A supermarket finds that customers who buy diapers are also likely to buy beer on Thursday nights. This insight leads to placing diapers and beer nearby, increasing sales of both.

The goal is to generate **association rules** of the form $X \Rightarrow Y$, where X and Y are disjoint sets of items.

2 Support, Confidence, Frequent Itemsets, and the Apriori Property

Let $I = \{i_1, i_2, \ldots, i_n\}$ be the set of all items, and $D = \{t_1, t_2, \ldots, t_m\}$ a database of transactions, where each $t_j \subseteq I$.

Basic Measures

1. **Support** of an itemset S (or a rule $X \Rightarrow Y$):

$$\text{support}(S) = \frac{\text{number of transactions containing } S}{|D|}$$

   It measures the popularity of S. For a rule $X \Rightarrow Y$, $\text{support}(X \Rightarrow Y) = \text{support}(X \cup Y)$.

2. **Confidence** of a rule $X \Rightarrow Y$:

$$\text{conf}(X \Rightarrow Y) = \frac{\text{support}(X \cup Y)}{\text{support}(X)}$$

   It measures the conditional probability that a transaction containing X also contains Y.

3. **Frequent itemset**: An itemset whose support is at least a user-specified minimum support threshold min_sup.

The Apriori Property

The **Apriori property** states: *All non-empty subsets of a frequent itemset must also be frequent.* Conversely, if an itemset is infrequent, all its supersets are infrequent. This property is used to prune the search space during itemset generation.

Numerical Example (Running Example) Consider a small transaction database with 5 transactions:

Transaction ID	Items
T1	Milk (M), Bread (B), Diaper (D)
T2	Milk (M), Bread (B), Diaper (D), Cola (C)
T3	Milk (M), Diaper (D), Beer (E)
T4	Milk (M), Bread (B), Diaper (D), Beer (E)
T5	Diaper (D), Beer (E)

Let min_sup $= 0.6$ (i.e., at least $0.6 \times 5 = 3$ transactions). First, count single items:

- Milk: appears in T1,T2,T3,T4 $\Rightarrow$ count 4, support $4/5 = 0.8$
- Bread: T1,T2,T4 $\Rightarrow$ count 3, support 0.6
- Diaper: all 5 $\Rightarrow$ count 5, support 1.0
- Beer: T3,T4,T5 $\Rightarrow$ count 3, support 0.6
- Cola: only T2 $\Rightarrow$ count 1, support 0.2 (infrequent)

Thus frequent 1-itemsets (L1): $\{M\}, \{B\}, \{D\}, \{E\}$.

Next, generate candidate 2-itemsets from L1 and count:

- $\{M, B\}$: T1,T2,T4 $\Rightarrow$ 3 (frequent)
- $\{M, D\}$: T1,T2,T3,T4 $\Rightarrow$ 4
- $\{M, E\}$: T3,T4 $\Rightarrow$ 2 (infrequent)
- $\{B, D\}$: T1,T2,T4 $\Rightarrow$ 3
- $\{B, E\}$: T4 only $\Rightarrow$ 1 (infrequent)
- $\{D, E\}$: T3,T4,T5 $\Rightarrow$ 3

Frequent 2-itemsets (L2): $\{M, B\}, \{M, D\}, \{B, D\}, \{D, E\}$.

Now candidate 3-itemsets are built from L2. Only $\{M, B, D\}$ has all its 2-subsets in L2; count it: appears in T1,T2,T4 $\Rightarrow$ 3 (frequent). $\{M, D, E\}$ is not considered because $\{M, E\}$ is infrequent. No candidate 4-itemsets because $\{M, B, D, E\}$ has $\{M, E\}$ infrequent.

Thus the only frequent 3-itemset is $\{M, B, D\}$.

3 How Does the Apriori Algorithm Work?

The Apriori algorithm generates frequent itemsets through a level-wise iterative process, using the Apriori property to prune.

Algorithm 1 Apriori Algorithm

Input: Database D, min_sup
Output: Frequent itemsets L
$L_1 = \{\text{frequent 1-itemsets}\}$
for $k = 2; L_{k-1} \neq \emptyset; k++$ **do**
    $C_k = \text{apriori\_gen}(L_{k-1})$ ▷ Generate candidates
    for all transactions $t \in D$ **do**
        $C_t = \text{subset}(C_k, t)$ ▷ Candidates contained in t
        for all candidates $c \in C_t$ **do**
            $c.\text{count}++$
        end for
    end for
    $L_k = \{c \in C_k \mid c.\text{count} \geq \text{min\_sup\_count}\}$
end for
return $L = \bigcup_k L_k$

Step-by-Step Example (Continued)

From our running example:

1. L1 = $\{M\}, \{B\}, \{D\}, \{E\}$.
2. Generate C2 from L1: all pairs. Prune using support counts; we get L2 as above.
3. Generate C3: join $\{M, B\}$ and $\{M, D\}$ to get $\{M, B, D\}$; its subsets $\{M, B\}, \{M, D\}, \{B, D\}$ are all frequent, so keep. Similarly $\{B, D, E\}$? Subset $\{B, E\}$ is infrequent, so discard.
4. Count $\{M, B, D\} = 3$, so L3 includes it.
5. No further candidates.

Once frequent itemsets are found, rules are generated by partitioning each frequent itemset into antecedent and consequent. For each rule, we compute confidence and keep those above a threshold (e.g., min_conf = 0.75).

Rule Generation from $\{M, B, D\}$

Frequent itemset $\{M, B, D\}$ has 3 transactions. Possible rules (with confidence):

- $\{B, D\} \Rightarrow \{M\}$: conf = 3/3 = 100% (strong)
- $\{M, B\} \Rightarrow \{D\}$: conf = 3/3 = 100% (trivial)
- $\{M, D\} \Rightarrow \{B\}$: conf = 3/4 = 75% (meets threshold)
- $\{D\} \Rightarrow \{M, B\}$: conf = 3/5 = 60% (pruned)
- $\{M\} \Rightarrow \{B, D\}$: conf = 3/4 = 75% (strong)

Thus we obtain several actionable rules.

4 Extension from Flag Data to General Categorical Data

The standard association rule formulation assumes *flag data* (presence/absence of items). However, real datasets often contain categorical attributes with multiple values. We can extend the approach by:

- **Itemization**: Convert each attribute–value pair into a separate binary item. For example, attribute *Age* with values {Young, Middle, Old} becomes three items: Age_Young, Age_Middle, Age_Old.
- **Taxonomies**: Use hierarchies (e.g., product categories) to mine rules at different levels of granularity.
- **Numerical attributes**: Discretize into intervals (e.g., Income_High, Income_Low).

Numerical Example: Categorical Data Consider a customer dataset with two attributes: *Age* and *Marital Status*. Transactions become:

Trans. ID	Age	Marital Status
1	Young	Single
2	Young	Married
3	Middle	Married
4	Old	Married
5	Old	Single

After itemization, we have binary items: Age_Young, Age_Middle, Age_Old, Marital_Single, Marital_Married. Then we can mine rules like:

$$\text{Age\_Young} \Rightarrow \text{Marital\_Single} \quad (\text{support} = 0.4, \text{confidence} = 0.667)$$

This allows association mining on any relational data.

5 Information-Theoretic Approach: Generalized Rule Induction Method

Traditional measures (support, confidence) may generate many trivial or redundant rules. Information-theoretic measures evaluate the quality of a rule by quantifying the reduction in uncertainty.

One popular measure is the **J-measure** (or cross-entropy), defined for a rule $X \Rightarrow Y$:

$$J(X \Rightarrow Y) = P(X) \cdot \left[P(Y|X) \log \frac{P(Y|X)}{P(Y)} + (1 - P(Y|X)) \log \frac{1 - P(Y|X)}{1 - P(Y)} \right]$$

It balances the generality of the antecedent (via $P(X)$) and the information gain of the rule. A high J-measure indicates a rule that is both surprising and accurate.

Example From our running data, consider the rule $\{B, D\} \Rightarrow \{M\}$:

- $P(B, D) = 3/5 = 0.6$, $P(M|B, D) = 1.0$, $P(M) = 4/5 = 0.8$.
- $J = 0.6 \times [1.0 \cdot \log(1.0/0.8) + 0 \cdot \log(0/0.2)] = 0.6 \times \log(1.25) \approx 0.6 \times 0.223 = 0.134$ (using natural log).

The J-measure helps rank rules beyond simple confidence.

6 Association Rules Are Easy to Do Badly

While association rule mining is powerful, it is also easy to produce misleading results. Common pitfalls include:

1. **Spurious correlations**: With low support, rules may appear significant by chance. For example, in a huge transaction database, a rule like "Buying coffee $\Rightarrow$ buying toothpaste" may have high confidence but low support, yet no causal link exists.
2. **Too many rules**: Setting low thresholds can generate millions of rules, overwhelming the analyst.
3. **Redundant rules**: Rules with the same information content (e.g., $A \Rightarrow B$ and $A \Rightarrow B, C$) may be redundant.
4. **Confidence alone is insufficient**: A rule with high confidence may still be uninteresting if the consequent is already very frequent (e.g., "Buying milk $\Rightarrow$ buying bread" when bread is bought 90% of the time).
5. **Ignores negative patterns**: Traditional association rules only capture positive co-occurrence, missing interesting negative relationships (e.g., "if customer buys diet soda, they do not buy regular soda").

7 How Can We Measure the Usefulness of Association Rules?

Beyond support and confidence, several measures help assess rule utility:

1. **Lift** (interest factor):
$$\text{Lift}(X \Rightarrow Y) = \frac{\text{conf}(X \Rightarrow Y)}{\text{support}(Y)} = \frac{P(X \cup Y)}{P(X)P(Y)}$$
Lift > 1 indicates positive correlation; $= 1$ independence; < 1 negative correlation.
2. **Conviction**:
$$\text{Conviction}(X \Rightarrow Y) = \frac{1 - \text{support}(Y)}{1 - \text{conf}(X \Rightarrow Y)}$$
Measures how much Y depends on X. Values > 1 indicate a strong association; infinite when confidence $= 1$.
3. **Leverage**:
$$\text{Leverage}(X \Rightarrow Y) = \text{support}(X \cup Y) - \text{support}(X) \cdot \text{support}(Y)$$
Measures the difference between observed and expected co-occurrence.
4. **Chi-square test**: Tests statistical independence between X and Y.

Numerical Example (Lift) For rule $\{B, D\} \Rightarrow \{M\}$ from earlier:

$\text{Support}(\{B, D\} \cup \{M\}) = 3/5 = 0.6, \quad \text{Support}(\{B, D\}) = 3/5 = 0.6, \quad \text{Support}(\{M\}) = 4/5 = 0.8.$

$$\text{Lift} = \frac{0.6}{0.6 \times 0.8} = \frac{0.6}{0.48} = 1.25.$$

Since lift > 1, the rule is positively useful.

8 Do Association Rules Represent Supervised or Unsupervised Learning?

Association rule mining is generally considered an **unsupervised** learning method because:

- It does not require a predefined target variable.
- It discovers patterns based on co-occurrence without labeled classes.
- The rules are symmetric in the sense that the antecedent and consequent are not predetermined.

However, there are variants where the consequent is fixed (e.g., finding rules that predict a particular product), which can be seen as a form of supervised learning. In practice, association rule mining is used for exploratory analysis to uncover hidden structures, aligning it with unsupervised tasks.

9 Local Patterns versus Global Models

Association rules are **local patterns** that describe specific relationships among subsets of the data. They differ from **global models** (e.g., regression, decision trees, neural networks) in several ways:

- **Scope**: Local patterns apply to a subset of the data (the transactions that satisfy the antecedent), while global models attempt to capture overall behavior across all instances.
- **Interpretability**: Rules are often easy to interpret and can be used for targeted actions (e.g., a specific promotion for a segment).
- **Redundancy**: Many local patterns may overlap; a global model summarizes the data more compactly.
- **Generality**: Global models are better suited for prediction on new data, whereas local patterns are descriptive.

Example A global model might say: "Customers who spend more than $100 have a 70% probability of being loyal." A local pattern (rule) might say: "If a customer buys diapers and beer, they also buy chips in 80% of cases." The rule provides a specific, actionable insight for a narrow scenario, complementing the broader view of a global model.

10 Conclusion

Association rule mining provides a powerful framework for discovering hidden relationships in large datasets. By understanding the key measures (support, confidence, lift), the Apriori algorithm, and extensions to categorical and information-theoretic approaches, analysts can extract valuable insights. However, care must be taken to avoid common pitfalls and to evaluate rules using multiple usefulness measures. Ultimately, association rules serve as local, unsupervised patterns that complement global modeling techniques.